



MALVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Advanced Semiconductor Devices (21ECT541)

Mid-Semester Examination: 27th September 2024

Time Allowed – 1:30 Hrs

MM-30

Notes:

- This question paper contains 6 questions.
- Marks are indicated against each question.
- Question no. 01 is compulsory.
- Attempt any four questions between Q. No. 02, 03, 04, 05 & 06.

Course Outcomes:

At the end of the course the student will be able to:

- CO1 Analyse MOSFET functionalities and physics
- CO2 Understand Short Channel Effects (SCEs)
- CO3 Design Scaled MOSFETs.
- CO4 MOSFET Modeling for scaled devices.
- CO5 Familiarisation with state of art semiconductor devices.

S. No.	Question Description	Marks	Mapped Course Outcome
1.	Differentiate between following (Briefly): a) Direct Band Gap Semiconductors v/s Indirect Band Gap Semiconductors b) Intrinsic v/s Doped Semiconductors c) HEMTs v/s MESFETs d) MOSFET v/s BJT e) Long Channel v/s Short Channel Transistors f) MOSFET as a Voltage Variable Resistor v/s MOSFET as an Amplifier g) Rectifying v/s Ohmic contact	2*7=14 Marks	CO1, CO2, CO3 & CO4
2.	Explain the structure and fundamental operation of MESFET. How this is different from MOSFET?	04 Marks	CO1, CO2 & CO5
3.	Explain CMOS fabrication steps and briefly define Latch-up problem in CMOS logic design.	04 Marks	CO2, CO3 & CO4

<u>4.</u>	Explain the fundamental operation of HEMT and it's applications.	04 Marks	CO1, CO2, & CO5
<u>5.</u>	Explain C-V Characteristics of MOSFET and it's variation with respect to frequency.	04 Marks	CO1 & CO2
<u>6.</u>	Explain various applications of Nanotechnology in the field of Electronics.	04 Marks	CO5

Malaviya National Institute of Technology, Jaipur

Materials Research Centre

Mid Semester Examination, B. Tech 4th Year (Odd Semester)

Paper Name: Introduction to Soft Materials (21MST805) (Open elective)

Total Marks: 30

Time: 90 minutes

Date: 26/09/2024

Answer all the Questions.

1. What do you mean by the terms dispersion, suspension, and solution? (3 marks).
2. Explain the process of nanoprecipitation with suitable example? Why is the size of colloids measured using dynamic light scattering (DLS) greater than that measured using transmission electron microscopy (TEM)? (2+2=4).
3. What is Brownian motion in colloids? What are the conditions for choosing perfect particles for studying Brownian motion? Using a suitable example, explain the concept of electrical double layer (1+3+3 = 7 marks).
4. Calculate the fractional contribution of Debye, Keesom and London dispersion forces for *t*-butanol from the given data at 25 °C. (6 marks)
 μ (Debye) = 1.67 $\frac{\alpha \times 10^{-30}}{4\pi\epsilon}$ (m³) = 9.46 $\beta \times 10^{77}$ (Jm⁶) = 5.46
5. Although ΔH for micelle formation is positive, micelles do form. Explain? What are the factors that affect CMC. (2+3 = 5 marks)
6. How does the CMC affect the turbidity and molar conductivity of surfactants? Show that the CPP of surfactants in spherical micelles is equal to or less than 1/3. (3+2=5 marks).

3.33 x 10³⁰

$\frac{k_B T}{a}$

50.97 x 100
~~2.58 x 10³⁰~~
50.97 x 10



$\frac{V_s}{V_c}$
 $\frac{V_s}{V_c}$
 $\frac{V_s}{V_c}$

Malaviya National Institute of Technology Jaipur
Department of Physics
 Mid term Examination (Odd Semester 2024-25)

Course Name: ITR and Cosmology
 Course Code: PHT-415
 Class : B.Tech (Open Elective)

Max.Marks:30
 Time:1.5 hrs
 Date:25/09/2024

Answer all Questions

1. Derive an expression for the relativistic addition of velocities for the Components of u_x , u_y and u_z ? [6M]
2. Derive the Lorentz transformations? When the Lorentz transformations reduces to Galilean transformations? [6M]
3. Describe the Michaelson-Morley experiment with neat diagram and explain the results of it? [6M]
4. A Spaceship is moving at $0.6c$ relative to Earth. A Second spaceship is moving at $0.7c$ relative to the first spaceship. What is the velocity of the second spaceship as observed from Earth? [6M]
5. A rod of proper length 20 meters is moving at $0.95c$ relative to a stationary observer. The rod is aligned at an angle of 30° with respect to its direction of motion (not parallel to the motion). What is the observed length of the rod as measured by the stationary observer [6M]

End

Handwritten notes and diagrams:

Diagram 1: Shows two spaceships moving relative to Earth (S). One moves at $0.6c$ and the other at $0.7c$.

Diagram 2: Shows a rod of length $20m$ moving at $0.95c$ relative to a stationary observer (S). The rod is at an angle of 30° to its direction of motion.

Equations and formulas:

- $u_x = 0.6c$
- $x_1 = \gamma(x'_1 + vt'_1)$
- $x_2 = \gamma(x'_2 + vt'_2)$
- $L = \gamma(L_0)$
- $L > L_0$
- $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$
- $v \cos 30^\circ$
- $v \sin 30^\circ$
- $L = \gamma L_0$ (moving)
- $\cos 30^\circ = \frac{\sqrt{3}}{2}$
- $L_1 = \frac{L_0}{\sqrt{1 - \frac{v^2}{c^2}}}$



Roll No. _____

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY, JAIPUR

Department of Electronics & Communication Engineering

Mid-Term Examination, September 2024

B.Tech 7th Semester

Subject: Satellite Communication & Radar Engineering (ECT670)

Time Allowed – 1:30 hrs.

MM-30

Important Instructions:

- Write your Roll No. on the space provided in the question paper.
- This question paper contains 5 questions. All questions are compulsory.
- Figures to the right indicate maximum marks.

- Q: 1** How satellite orbit are defined? What is the difference between geosynchronous and a geostationary satellite? **6**
- Q: 2** List out the frequency bands used for satellite services. **6**
- Q: 3** State and explain Kepler's laws. Write down the Kepler's orbital elements? **6**
- Q: 4** What are the look angles? Explain how look angles are determined using sub-satellite points? Derive the necessary expressions for look angles. **6**
- Q: 5** A Satellite in an elliptical orbit around the earth has an apogee 41174 km and a perigee of 550 km. What is the orbital period of this satellite? Give answer in hours. (Note: Radius of the earth is 6,378.137 km Kepler's Constant = $3.9860004418 \times 10^5 \text{ km}^3/\text{s}^2$) **6**

Handwritten solution for Q: 5:

$$\mu = G M_E$$
$$G M_E = \frac{4\pi^2 a^3}{T^2}$$
$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$
$$T = \frac{2\pi (41174 + 550)^{3/2}}{\sqrt{3.9860004418 \times 10^5}}$$
$$T = 20.27 \text{ sec.}$$



Malaviya National Institute of Technology Jaipur
Department of Electronics and Communication Engineering

Mid-Term exam
VII Semester B.Tech.

ECT620 Microelectronic Devices and Circuits

September 23, 2024

Max marks: 30

Instructions:

1. Interpret any bad data (if any).
2. Attempt all questions.

Student ID:

1. (a) Explain how the band formation takes place with an orbital diagram also elaborate how it happens in direct and indirect band semiconductor materials? (3)
(b) What is the fermi-energy level and how does the position of fermi level varies in intrinsic and extrinsic semiconductors with mathematical formulae. (3)
2. (a) What are high-frequency devices? Discuss the properties of VMOS and DMOS devices with their structure. (3)
(b) What are the parasitic effects in MOS devices? Draw the energy band diagram of the metal- semiconductor junction of n-type semiconductor. (3)
3. (a) What are interface effects and subthreshold region in MOS devices, and how do the interface trap effects influence the electrical parameters of the FET device? (3)
(b) What are the MOS Capacitances? Draw the capacitance vs gate voltage characteristics of a simple MOS capacitor at low frequency. (3)
4. (a) Silicon is doped with boron to a concentration of 2×10^{17} atoms/cm³. Assume the intrinsic carrier concentration of silicon to be 1.5×10^{10} /cm³ and the value of kT/q to be 25 mV at 300 K. Compared to undoped silicon, the fermi level of doped silicon goes ___ by ___ eV. (3)
(b) Explain the working of N-MOSFET in depletion mode and CMOS Inverter with voltage transfer characteristic curve. (3)
5. What are the HEMTs and MESFETs devices, explain their working operation. (6)

$$f(r) = \ln \left(\frac{N_A N_D}{n_i^2} \right) \left(\frac{kT}{2q} \right)$$

25mV = $\ln$
Linear Saturation dep
inversion



Time Allowed – 2:30 Hrs

MM-50

Notes:

- This question paper contains 10 questions.
- Marks are indicated against each question.

Course Outcomes:

At the end of the course the student will be able to:

- CO1 Analyse MOSFET functionalities and physics
- CO2 Understand Short Channel Effects (SCEs)
- CO3 Design Scaled MOSFETs.
- CO4 MOSFET Modeling for scaled devices.
- CO5 Familiarisation with state of art semiconductor devices.

S. No.	Question Description	Marks	Mapped Course Outcome
1.	Explain the fundamental operation of Junctionless transistor and its applications.	05 Marks	CO3, CO4 & CO5
2.	Define all types of leakage problems in conventional MOSFETs.	05 Marks	CO1, CO2, & CO3
3.	Describe electrical nature of MS contact with respect to type of semiconductor material and work function of the material.	05 Marks	CO2 & CO3
4.	Explain the fundamental operation of HEMT and it's applications.	05 Marks	CO1, CO2, & CO3
5.	Explain short channel effects in detail.	05 Marks	CO1 & CO2
6.	What is SOI technology and explain its advantages and limitations over conventional MOSFETs.	05 Marks	CO3, CO4 & CO5

71/11/24

7.	Explain various applications of Nanotechnology.	05 Marks	CO3
8.	Explain various structures of Tunnel FET and its fundamental operation.	05 Marks	CO3, CO4 & CO5
9.	Explain the fundamental operation of FinFET and its applications.	05 Marks	CO3, CO4 & CO5
10.	Define process corners and environmental variations for non-ideal transistors.	05 Marks	CO1, CO2, & CO3